

Assignment 1 — Pima Indians Diabetes

EDA & CLASSIFICATION REPORT

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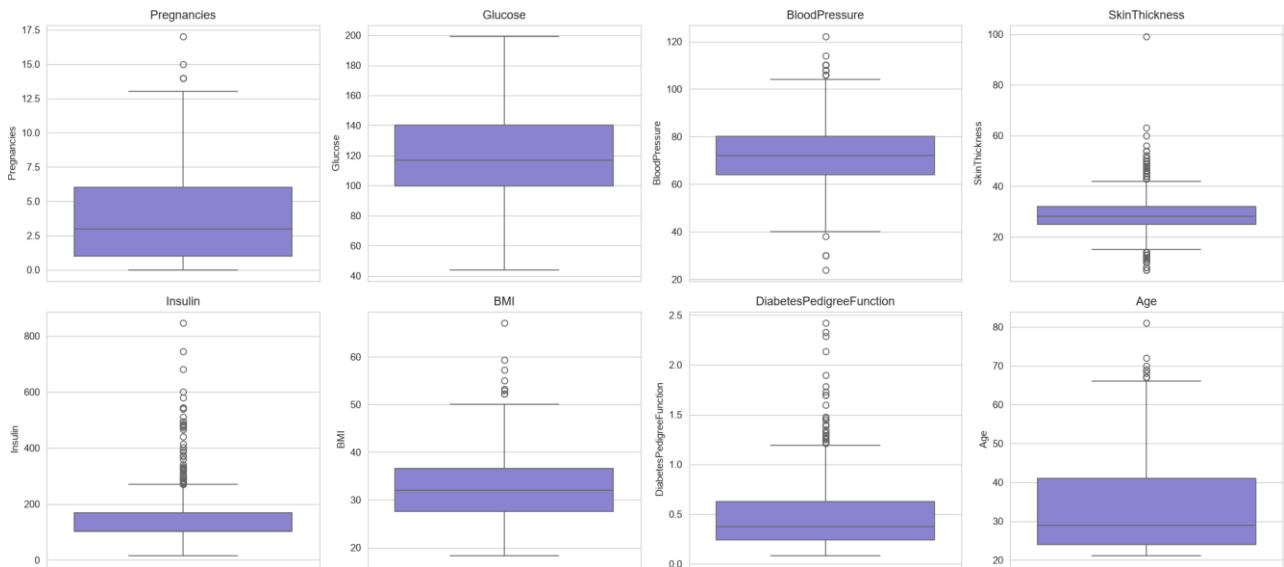
Dataset shape: 768 rows × 9 columns

1. Data Quality Report

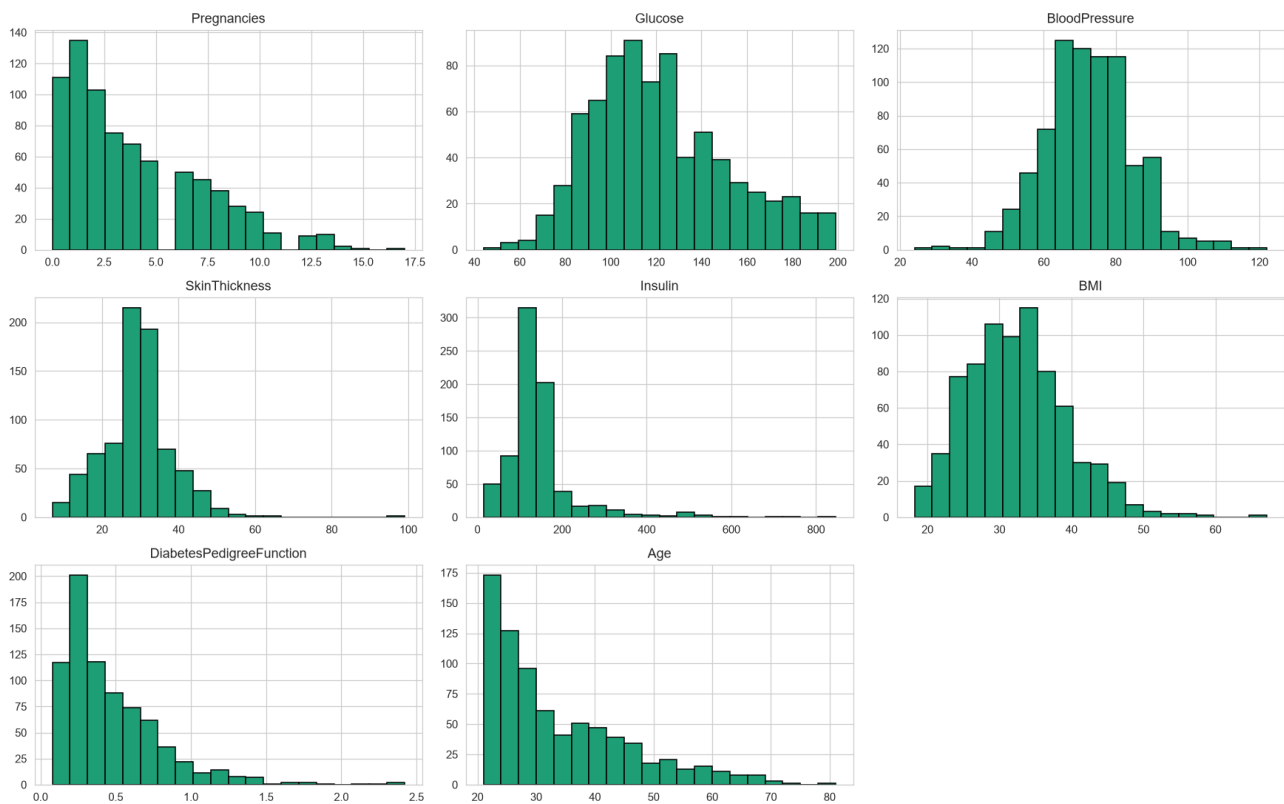
	dtype	cardinality
Pregnancies	int64	17
Glucose	float64	135
BloodPressure	float64	47
SkinThickness	float64	50
Insulin	float64	187
BMI	float64	247
DiabetesPedigreeFunction	float64	517
Age	int64	52
Outcome	int64	2

2. Visualizations

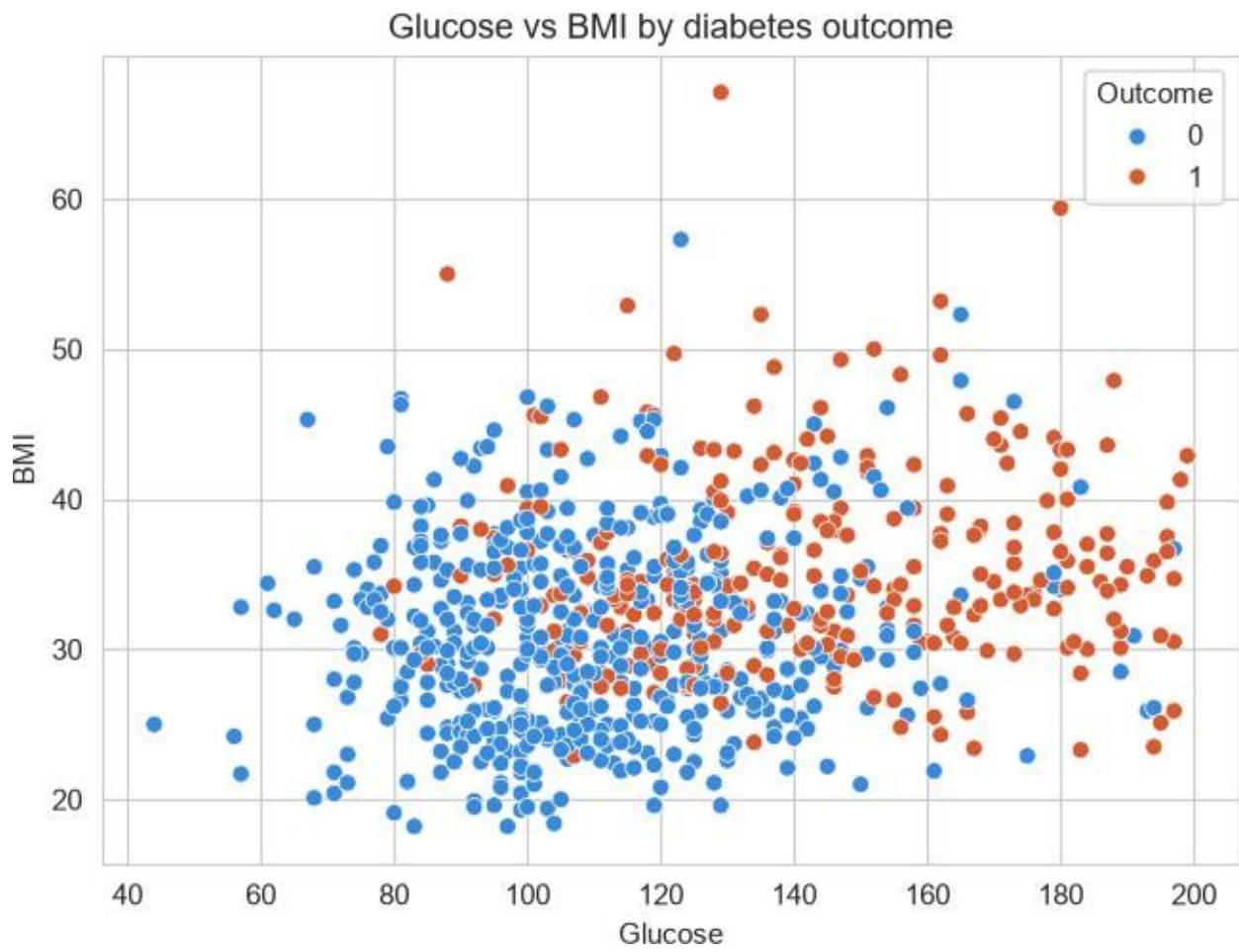
Boxplots



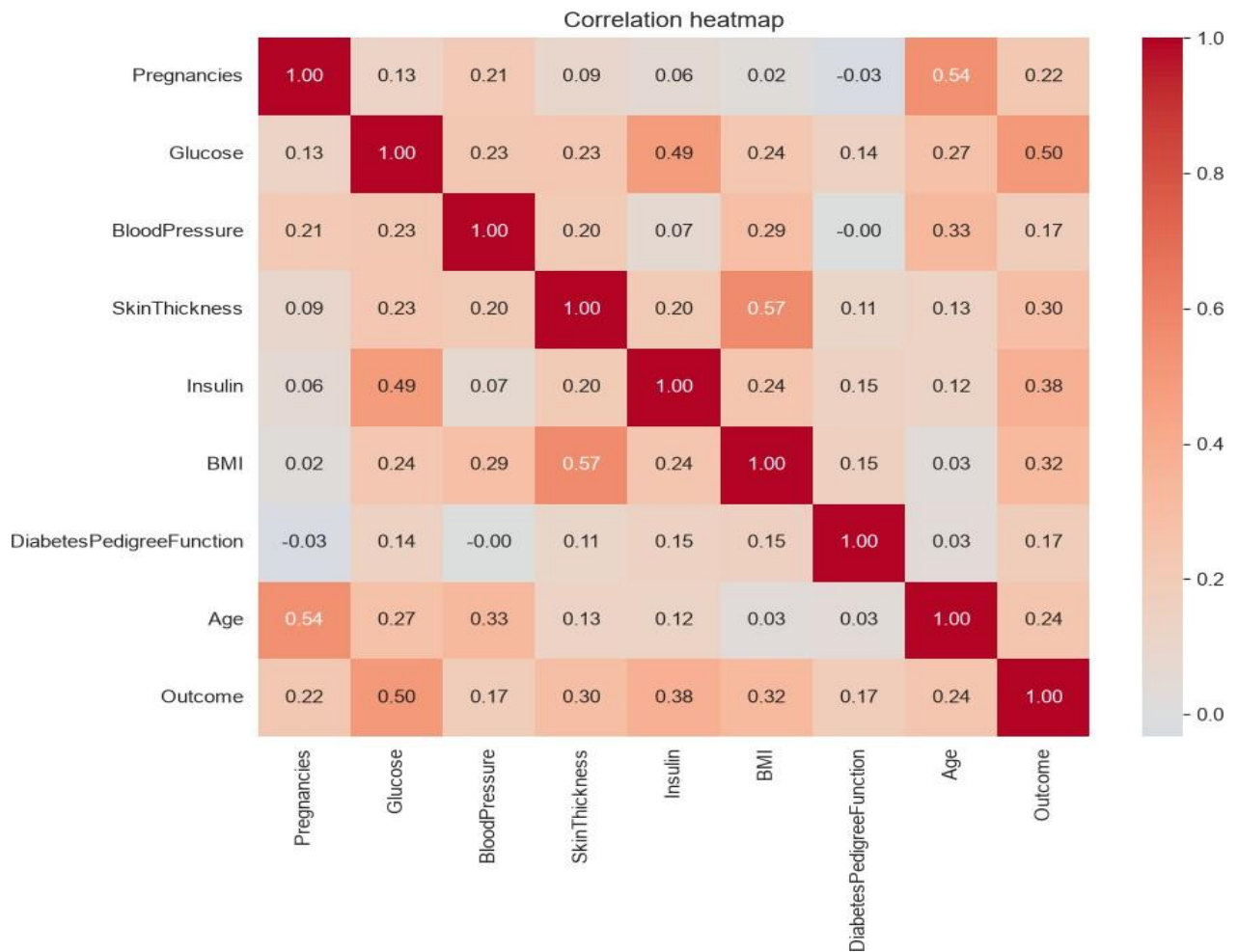
Histograms



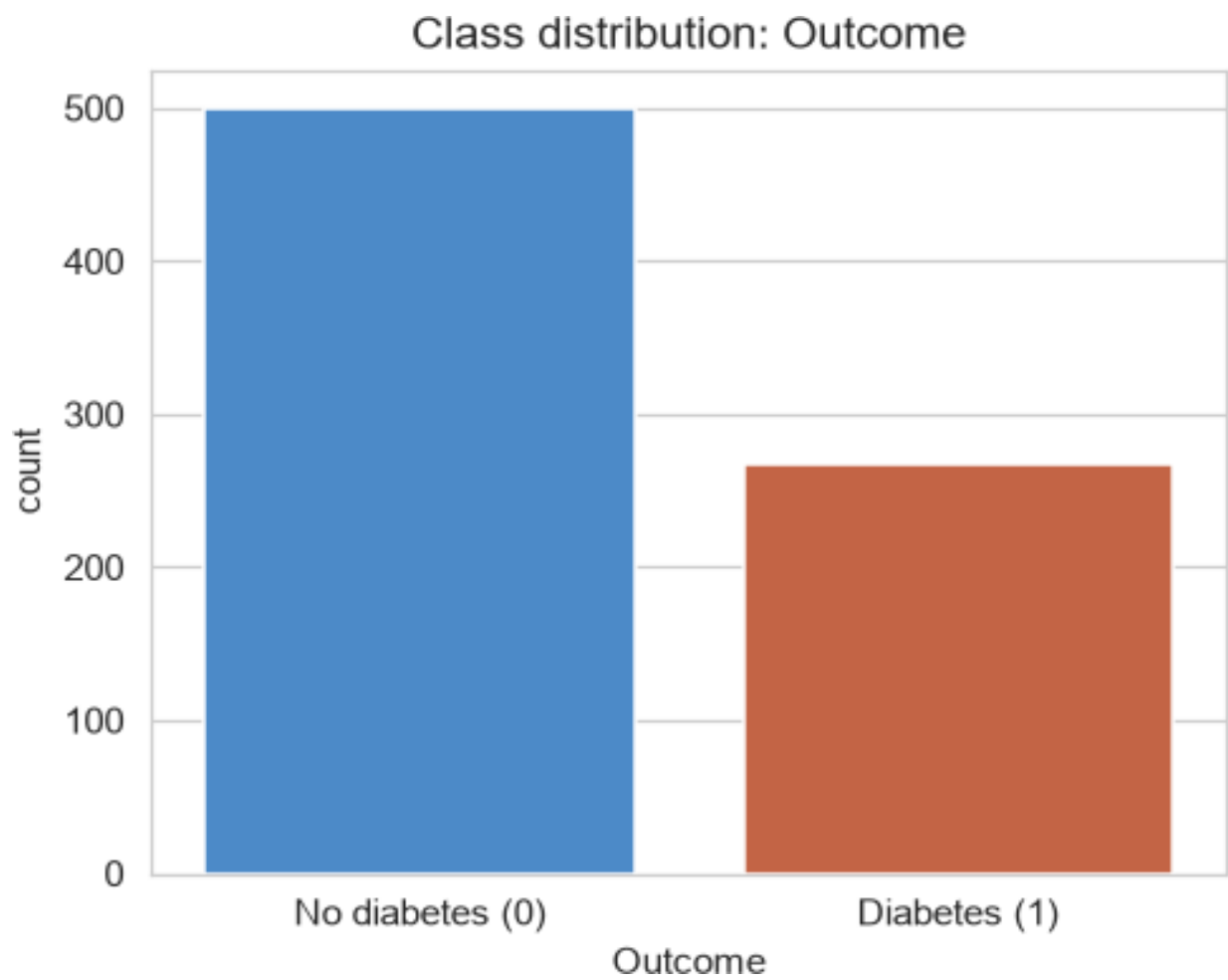
Scatter: Glucose vs BMI



Correlation heatmap



Class Distribution



3. Model Results

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
KNN	0.773	0.638	0.815	0.715	0.844
Naive Bayes	0.734	0.6	0.722	0.655	0.813